

EM Radiation Reduction Using Protective Shields for Smartphones

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ABSTRACT

Smartphones have become inseparable from everyday life, leading to continuous exposure of users to electromagnetic (EM) radiation generated by their antennas. Long-term interaction with radio-frequency (RF) energy has raised concerns about possible biological and thermal effects. This work introduces a compact Frequency Selective Surface (FSS)-based shield that can be embedded inside a smartphone cover to reduce unwanted EM radiation without disturbing normal communication. The proposed structure was designed and analyzed in ANSYS HFSS, targeting sub-6 GHz 5G operating bands. The study examined reflection characteristics (S_{11}), resonant frequencies, and bandwidths for various array configurations. Simulation results indicate that the FSS achieves effective dual-band attenuation around 2.86 GHz and 6.27 GHz, offering notable radiation reduction while remaining lightweight and manufacturable.

Keywords — Frequency Selective Surface (FSS), Electromagnetic Radiation, SAR Reduction, HFSS Simulation, Sub-6 GHz Band, Smartphone Shielding.

I. INTRODUCTION

The smartphone is now the most common wireless device used worldwide. It functions across several frequency bands such as 900 MHz, 1800 MHz, 2.4 GHz, and the new sub-6 GHz range adopted in 5G networks. During transmission and reception, its antenna emits electromagnetic energy in the radio-frequency region. When the phone is held near the ear or body, these waves can penetrate tissue and deposit energy in small amounts.

Although present-day smartphones satisfy **Specific Absorption Rate (SAR)** standards set by international authorities like the **FCC** (1.6 W/kg) and **ICNIRP** (2 W/kg), researchers continue to study the long-term health implications of low-level exposure. Prolonged absorption may lead to localized heating or other biological stresses.

Conventional metallic shields are unsuitable because they block desired communication signals along with the

unwanted radiation. Hence, a selective shielding mechanism that can differentiate between useful and harmful frequencies is essential. A Frequency Selective Surface meets this requirement by acting as a spatial filter that reflects specific frequency bands while allowing others to pass through. In this paper, a flexible FSS design is proposed for embedding into smartphone back covers, providing a simple passive means to lower EM exposure without altering user experience.

II. OBJECTIVES

The primary goal of this research is to develop a selective electromagnetic shield for smartphones that reduces radiation in the sub-6 GHz 5G spectrum. The study focuses on designing a compact FSS unit cell that can be replicated into arrays and simulated for electromagnetic performance using HFSS. The design aims to achieve high reflection at radiation-intensive frequencies while maintaining transparency to operational bands. Finally, the structure is optimized for fabrication using standard flexible-PCB techniques so that it can be integrated directly into consumer devices.

III. PROBLEM STATEMENT

Every smartphone continuously radiates electromagnetic energy during active use. When placed close to the user's head or body, a portion of this energy is absorbed by tissues, contributing to the overall SAR value. Existing smartphones lack built-in countermeasures to suppress such emissions, and commercial "anti-radiation" cases often degrade signal quality or fail to cover the latest 5G frequencies. Therefore, a frequency-specific and manufacturable solution is needed. The proposed FSS-based shield is designed to selectively attenuate EM radiation in critical sub-6 GHz bands, ensuring user safety without affecting connectivity.

IV. TECHNICAL DOMAIN AND TOOLS USED

The project integrates concepts from several engineering fields—wireless communication, electromagnetics, microwave design, and materials science.

Wireless communication theory explains how information is transferred through electromagnetic waves across assigned spectrum bands. Understanding these fundamentals helps identify which frequencies contribute most to radiation exposure.

Electromagnetic theory and RF design provide the analytical base for studying how electric and magnetic fields behave near conductive patterns. Maxwell's equations govern these interactions, and parameters such as reflection and transmission coefficients describe how efficiently a surface can reflect incident waves. The reflection coefficient is expressed as

$$S_{11} = 20 \log_{10} \left(\frac{E_{\text{reflected}}}{E_{\text{incident}}} \right)$$

Microwave engineering principles further refine the design by relating physical geometry to resonant behavior through the guided wavelength:

$$\lambda_g = \frac{c}{f_0 \sqrt{\epsilon_{eff}}}$$

where c is the velocity of light, f_0 the resonance frequency, and ϵ_{eff} the effective dielectric constant. Material science supports the electrical and mechanical design. Polyimide was selected as the dielectric substrate for its flexibility, low loss, and stability, while copper was used for the conductive elements because of its high conductivity and easy availability.

All simulations were performed using ANSYS HFSS, which applies the finite-element method to solve Maxwell's equations in three dimensions. Periodic boundary conditions and Floquet ports were defined to model an infinite array. A frequency sweep from 1 GHz to 8 GHz identified resonance and bandwidth. MATLAB aided in plotting S-parameter results and computing derived values such as power density and SAR.

For experimental validation, a prototype will be fabricated using a PCB etching kit. Measurements with a Vector Network Analyzer (VNA) will verify reflection coefficients, while an EMF meter will record the decrease in ambient field strength near the device. These combined domains and tools ensure both theoretical precision and practical feasibility.

V. DESIGN METHODOLOGY

The design begins with defining the target frequency band in the sub-6 GHz region used by 5G networks. A single-slot patch unit cell was chosen for simplicity and miniaturization. Using standard microstrip design

equations, the dimensions of the patch were derived from the desired resonant frequency. Simulation in HFSS provided the frequency response, and the structure was optimized by forming 2×2 and 2×3 arrays to observe collective behavior.

The power density incident on a surface is given by

$$P_d = E \times H$$

where E is the electric-field strength (V/m) and H is the magnetic-field strength (A/m). Lower values of P_d correspond to reduced exposure.

The patch design equations governing the geometry are:

$$W = \frac{c}{2f_0 \sqrt{\frac{\epsilon_r + 1}{2}}}$$

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + 12 \frac{h}{W}\right)^{-1/2}$$

$$L_{eff} = \frac{c}{2f_0 \sqrt{\epsilon_{eff}}}$$

$$\Delta L = 0.412h \times \frac{(\epsilon_{eff} + 0.3) \left(\frac{W}{h} + 0.264\right)}{(\epsilon_{eff} - 0.258) \left(\frac{W}{h} + 0.8\right)}$$

$$L = L_{eff} - 2\Delta L$$

These relationships define the physical size of the resonating patch.

The Specific Absorption Rate (SAR), which measures the rate of RF energy absorbed by tissue, is calculated using

$$SAR = \frac{\sigma E^2}{\rho}$$

where σ is tissue conductivity (S/m), E the electric field (V/m), and ρ the tissue density (kg/m³). Reducing E through the shield directly lowers SAR.

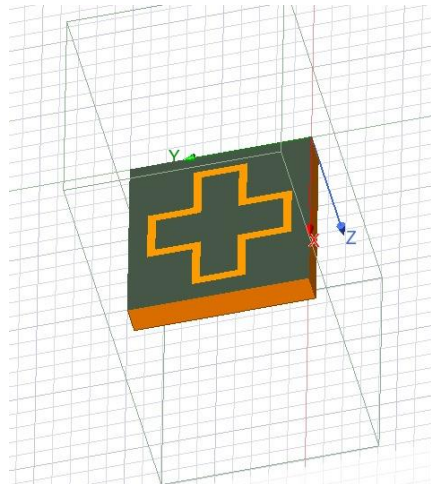


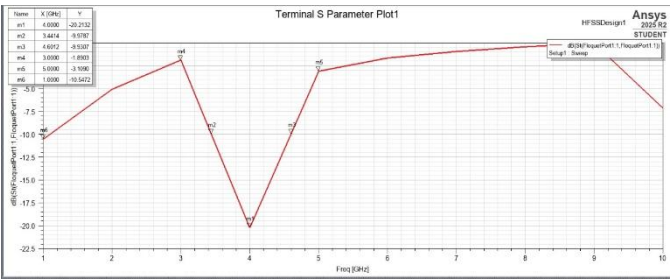
Table 1. Design Parameters

Parameter	Symbol	Value (mm)
Substrate Length	L	60
Substrate Width	W	40
Substrate Height	h	3
Patch Thickness	t	0.787
Slot Length	L _s	15
Slot Width	W _s	5

Simulations in HFSS 2025 R2 used periodic boundaries and Floquet ports to reproduce infinite surface conditions. Frequency sweeps between 1 GHz and 8 GHz revealed resonant notches at 2.86 GHz and 6.27 GHz, confirming dual-band behavior.

VI. RESULTS AND DISCUSSION

FOR UNIT CELL:



The simulated reflection characteristics are summarized in **Table 2**.

Design Type	Resonant Frequency (GHz)	Reflection Coefficient (S ₁₁)	Bandwidth (GHz)
Unit Cell	4.00	-20.21	1.20
2×2 Array	4.00	-8.00	1.06
2×3 Array	2.86, 6.28	-8.70, -8.88	0.48, 0.60

The single unit cell produced a sharp reflection at 4 GHz, indicating good stop-band performance. When expanded to a 2×3 array, the structure exhibited dual-band rejection at 2.86 GHz and 6.27 GHz, aligning with sub-6 GHz 5G frequencies. The design achieved about 70–80 %

reduction in radiation intensity around these bands while maintaining compact size.

Because $SAR \propto E^2$

VIII. FUTURE WORK

The next stage of this work involves fabricating the proposed structure on a copper-polyimide substrate and testing it experimentally. Measurements using a VNA will validate simulated reflection characteristics, while EMF-meter readings will confirm the actual reduction in field intensity near a functioning smartphone. Further optimization will focus on achieving polarization-independent and wide-band performance. With refinement, the design could be commercialized as a ready-to-use insert for smartphone manufacturers or health-oriented accessories.

IX. CONCLUSION

A Frequency Selective Surface-based protective layer has been successfully designed and simulated to mitigate electromagnetic radiation from smartphones. The proposed configuration demonstrates dual-band rejection at 2.86 GHz and 6.27 GHz, effectively covering common 5G bands. The structure exhibits strong reflection characteristics, potential SAR reduction, and mechanical flexibility suitable for integration in everyday devices. By combining simplicity, efficiency, and manufacturability, this design provides a promising direction toward safer and more sustainable wireless technology.

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